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Highland Games: A benchmarking exercise in predicting biophysical and drug properties of monoclonal antibodies from amino acid sequences

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Abstract:

Biopharmaceutical product and process development does not yet take advantage of predictive computational modeling to nearly the degree seen in industries based on smaller molecules. To assess and advance progress in this area, spirited coopetition (mutually-beneficial collaboration between competitors) was successfully used to motivate industrial scientists to develop, share, and compare data and methods which would normally have remained confidential. The first “Highland Games” competition was held in conjunction with the October 2018 Recovery of Biological Products Conference in Ashville, NC, USA, with the goal of benchmarking and assessment of the ability to predict development-related properties of six antibodies from their amino acid sequences alone. Predictions included purification-influencing properties such as isoelectric point and Protein A elution pH, and biophysical properties such as stability and viscosity at very high concentrations. Essential contributions were made by a large variety of individuals, including companies which consented to provide antibody amino acid sequences and test materials, volunteers who undertook the preparation and experimental characterization of these materials, and prediction teams who attempted to predict antibody properties from sequence alone. Best practices were identified and shared, and areas in which the community excels at making predictions were identified, as well as areas presenting opportunities for considerable improvement. Predictions of isoelectric point and Protein A elution pH were especially good with all-prediction average errors of 0.2 and 1.6 pH unit, respectively, while predictions of some other properties were notably less good. This manuscript presents the events, methods, and results of the competition, and can serve as a

tutorial and as a reference for in-house benchmarking by others. Organizations vary in their policies concerning disclosure of methods, but most managements were very cooperative with the Highland Games exercise, and considerable insight into common and best practices is available from the contributed methods. The accumulated data set will serve as a benchmarking tool for further development of *in silico* prediction tools.

Keywords: chromatography, aggregation, purification, molecular simulation, developability, QSAR

Introduction

Predicting the behavior of monoclonal antibodies and other biopharmaceuticals under processing and formulation conditions was practically impossible not long ago, and is still very difficult. The benefits of predictive tools are now widely realized in industries based on simpler molecules, however, and efforts to predict developability-related properties of proteins are long-standing and increasingly successful. A comparison of predictive methods using data blinded to the predictors has not been performed, nor have methods been compared across the industry. This paucity of comparisons, especially from industry, may be due to the reluctance of industry scientists and their legal teams to share data on proprietary molecules. Here coopetition, a collaboration between competitors in pre-competitive areas for mutual benefit, was adopted as a motivating factor to engage industrial and academic predictors across the community. Coopetition has increasingly been used in education (Busch, 2014; Hamari et al., 2014; Nah et al., 2014). Here, it is demonstrated to further the understanding of molecular predictions of manufacturing and formulation behavior of biopharmaceuticals.

The Highland Games competition held in conjunction with the 2018 Recovery of Biological Products Conference was an industry-wide benchmarking exercise designed to assess and improve the ability to predict developability properties of monoclonal antibodies (chromatography, stability, viscosity) from their amino acid sequences. Improvements in this capability would greatly advance antibody candidate selection and process development in the high-throughput development pipelines now widely used. Six mAbs from terminated programs, biosimilar programs, and the NISTmAb standard (Schiel and Turner, 2018) were experimentally evaluated for developability-related properties using the same analytical methods. In parallel, prediction teams were asked to generate in silico prediction from only the amino acid sequences using their methods of choice.

The mAb characteristics to be predicted included behaviors that are directly relevant to large scale manufacturing and formulation: MabSelect Sure elution pH and pool volume, sodium chloride concentrations at elution from Capto Q anion-exchange chromatography and POROS 50 HS cation-exchange chromatography, viscosity of 160 g/L mAb, aggregation at pH 3.4, acidic species formation, isoelectric pH (pI), and pH 5.8 stability against high molecular weight species and fragment formation. The experimental data was generated by generous volunteers at Boehringer Ingelheim and Roche/Genentech.

Prediction teams used a very wide variety of methods, including statistical correlation techniques, Quantitative Structure-Activity Relationship (QSAR), homology modeling and molecular simulations. Many predictions were based on

the prior experience of the predicting organization, with varying degrees of complexity. For example, one organization predicted Protein A elution pH by titrating the medium and comparing against previous observations, with a modest correction for the acidity of the particular antibody. Principal component analysis was a useful tool, as was homology modeling, e.g., based on the structure of human IgG1 b12 (PDB: 1HZH). Another popular approach was QSAR modeling, which correlates the observed experimental data (e.g., protein retention) with (sometimes very many) specific physicochemical descriptors such as charge distribution or hydrophobic surface area. Models are first trained on in-house data sets, then used to predict unknown Highland Games mAbs (Woo et al., 2015).

2. Materials and Methods

Five monoclonal antibodies, mAbs 1-4 and mAb 5 (Rituxan) were produced, purified, and characterized by volunteers at Boehringer Ingelheim and Roche/Genentech. The mAb 6, NISTmAb (100 g/L) was obtained from the National Institute of Standards and Technology (NIST). The amino acid sequences of the antibodies appear in SI Table 1. The purification of these antibodies is described in sections 2.1 – 2.3. The experimental characterization is described in sections 2.4-2.6.

Protein A Chromatography (Event 1A/B)

MabSelect SuRe Protein A resin (GE Healthcare) was used on an AKTA Avant instrument (GE Healthcare) controlled by Unicorn software version 7.3. A 1.1 cm diameter column was packed according to the manufacturer's recommendations to a final bed height of *ca.* 20 cm (range 19 - 22 cm), and packing quality was tested according to the manufacturer's specifications. The column was equilibrated with phosphate-buffered saline (PBS) at pH 7.4 for 5 column volumes (CV), followed by loading with harvested cell culture fluid (HCCF) at 35 mg antibody per mL of resin. The resin was then subjected to a PBS wash (> 5 CV) followed by a wash

with 5 CV of 50 mM sodium acetate, pH 5.5 before step elution of the mAb using 50 mM sodium acetate, pH 3.4. Flow rate for MabSelect SuRe equilibration through elution was based on a residence time of 4 minutes. For a 20 cm column bed height, the linear flow rate was 240 cm/h. During elution, the fractions with UV absorbance between 100-100 mAU (for 2 mm path-length) were pooled, with a maximum collection volume of 10 CV. Following elution, the mAb pool from the MabSelect SuRe column was neutralized to pH 5.5 using 1 M Tris base. After completion of elution, the column was washed with at least 3 CV of 1 M acetic acid stripping solution, followed by cleaning-in-place using 0.2 N sodium hydroxide. For long-term storage, the column was washed and stored in 2% benzyl alcohol, 100 mM sodium phosphate, pH 7.2.

Capto Q Chromatography (Event 2)

Intermediate polishing of the MabSelect SuRe pool and characterization of AEX elution behavior was carried out using Capto Q resin (GE Healthcare) in bind/elute mode to reduce process-related impurities such as host cell proteins, DNA and endotoxins. A 1.1 cm diameter column was packed according to the manufacturer's recommendations to a final bed height of 20 cm (range 16 to 25 cm). Packing tests were performed as per the resin manufacturer's specifications. The pH of the MabSelect SuRe mAb pool was adjusted to pH 8.0 + 0.1, using 1 M tris or 1 M acetic acid. The pH-adjusted mAb pools were filtered through a 0.2 µm filter following any storage time greater than 6 hours. The Capto Q column was equilibrated with at least 8 CV of 50 mM Tris pH 8.0. The conditioned mAb pool was then applied to the column at a load density of 10 mg/mL of the Capto Q

resin. The loading flow rate was adjusted such that the residence time was at least 4 min. The column was washed with at least 5 CV of equilibration buffer, followed by a 20 CV linear gradient of 0-0.5 M sodium chloride in 50 mM Tris, pH 8.0. The method was set up to collect pools between 5-5 mAU (for 2 mm path-length). The column was regenerated with at least 3 CV of 50 mM Tris, 3M sodium chloride followed by cleaning-in-place using 1 N sodium hydroxide. For long-term storage, the column was washed with and stored in 0.1 N sodium hydroxide.

POROS 50 HS Chromatography (Event 3)

A final polishing step, the measurement of cation-exchange elution behavior was performed using POROS 50 HS resin (Thermo Fisher Scientific). A 1.1 cm diameter column was packed according to the resin manufacturer's recommendations to a final bed height of 20 cm + 5 cm. The column was equilibrated with 5 CV of 50 mM sodium acetate buffer, pH 5.0. The pH and conductivity of the AEX pool were adjusted to pH 5.0 + 0.1 and < 6.7 mS/cm respectively, using 1 M acetic acid, 1 M Tris base, or water for injection as appropriate. The conditioned AEX pool was then applied to the POROS 50 HS column at a load density of 10 mg/mL of POROS 50 HS. The loading flow rate was adjusted such that the residence time was at least 4 min. Following loading, the column was washed with 3 CV of 50 mM sodium acetate buffer, pH 5.0, then eluted with a linear gradient of 0-0.5 M sodium chloride in 50 mM sodium acetate, pH 5.0 over 20 CV. The product was pooled between 100-100 mAU (for 2 mm path-length), and the column was regenerated with 50 mM Tris, 3 M sodium chloride for at least 3 CV followed by cleaning-in-place using 1 N sodium hydroxide. For long term storage, the column was washed with and stored in 0.1 N

sodium hydroxide. The elution pool from the POROS 50 HS was concentrated and diafiltered into 20 mM histidine, 10 mM sodium chloride, pH 5.8. Concentration and diafiltration was carried out using an Amicon Ultra – 15 (Ultracel-30K, Millipore) and a centrifuge (Rotina 420R, Hettich). The diafiltered mAb was used in the Viscosity, Isoelectric point, and Aggregation stability measurements.

Stability by size-exclusion HPLC (Event 5, 8A/B)

The samples for the viscosity measurements were stored at 20°C while the samples for the pI, HMW and fragment content were stored at 19-22°C. The content of high molecular weight species and fragments was analyzed after 0, 1, 2, 4, 8, 16, and 32 days of storage. Analytical size-exclusion HPLC analysis was performed at ambient temperature on a TSK-Gel G3000SW_{XL} column, (7.8 mm ID × 300 mm H, 5 µm particle size; Cat. 08541, Tosoh Bioscience) on a Waters 2795 HPLC system equipped with UV detection at 280 nm (Waters 2489 UV/Vis detector) or an UltiMate 3000 HPLC system (ThermoFisher). On each system, elution at 0.5 mL/min was isocratic in 100 mM sodium phosphate, 200 mM L-arginine, 0.05% sodium azide at pH 6.8, or in 250 mM potassium chloride in 200 mM potassium phosphate, pH 7.0. 150 µg of protein was injected for each analysis.

Imaged Capillary Isoelectric Focusing with UV Detection (Event 6, 7)

The analysis used an iCE3 analyzer equipped with an Alcott 720 autosampler (ProteinSimple). For separation, a fluorocarbon-coated cartridge was used with 80 mM phosphoric acid in 0.1% methylcellulose (ProteinSimple) as anolyte and 100

mM sodium hydroxide in methylcellulose as catholyte. The reagent mix consisted of 0.35% methylcellulose and 2.5 mM (Roche) (or 1% methylcellulose and 3 mM arginine; Boehringer Ingelheim), 4% Pharmalyte 3-10 (17-0456-01 or 17-04563-01, GE Healthcare), 1% pI-markers (pI 5.85 and 9.46) and purified water. The final protein concentration in the reagent mix was 0.4 g/L. Samples were prefocused for 1 min at 1500 V, followed by a hold at 3000 V for 10 min. The main component and charge variants were focused to their respective pI values and then acquired using a whole-column UV detector at 280 nm.

Viscosity (Event 4)

Antibodies were concentrated using an Amicon[®] Ultra – 15 (Ultracel-30K, Millipore) filter and a Rotina 420R centrifuge (Hettich) operated at 4500 RPM (or 75 s⁻¹), corresponding to 3985x g. The procedure was repeated until the volume calculated to give 160 g/L was reached, or until the attempt was abandoned because the material could not be concentrated further. A Stabinger (Anton Paar) or microVISC (RheoSense) viscometer was used at 25°C.

Prediction Methods

3.1 Homology Modeling and Computation of Physicochemical Descriptors

A typical successful procedure for these central methods is as follows: Fab (Light and Heavy chain), forms the structural core of the designated binding domain. Once a framework Fab region is determined, the CDR loop regions are assigned by searching a database and assigning CDR loops based on loop length and similarity before grafting to the Fab dimer framework. To build a complete antibody, the Fab is superimposed on a full IgG structure of similar sequence and

grafted to build a full IgG homology model (Morea et al., 2000). The final grafted structure is energy-minimized to remove clashes and optimize the model. In one team's approach, the 3D structure of the Fv region of each mAb was first predicted from the primary sequence using Schrodinger's antibody modeling protocol (Zhu et al., 2014). The default parameters were used with the exception that the selection of templates for the heavy and light chain frameworks was not restricted to the same antibody structure.

Homology modeling by another successful team used MOE software (Chemical Computing Group ULC, 2016) to construct the Fab homology model structures from the sequence. Full-length antibody homology models used in prediction of viscosity and pI were generated by aligning (by alpha carbon RMSD minimization) the Fab models to either the left or right arm of a generic full-length IgG₁ molecule. The mAb IgG₁ framework was based on a full structure of the IgG₁ antibody that was pieced together from crystal structures of the fragments (Brandt et al., 2010). Although the sequences of the Highland Games mAbs contain C-terminal lysines, these are expected to be rapidly cleaved during the cell culture process. (Cai et al., 2011). Therefore, the C-terminal lysines were removed from the full-length structures. The Fab (or mAb) structures were energy-minimized using SANDER available in Case et al., 2015 to remove steric clashes. The relaxed structures (Fabs or mAbs) then were analyzed with PDB2PQR version 1.8 (Dolinsky et al., 2004) to determine the protonation states for ionizable residues at the respective pHs.

Biophysical descriptors were calculated for each of the homology model

structures at specific pHs (3.5 to 8.0, at 0.5 pH unit intervals) as required for the QSAR model building. The descriptors can be separated into two different categories; protein biophysical properties and protein surface properties. The protein biophysical properties were calculated for the entire molecular structure. The protein surface properties were calculated for the entire molecule and grouped under different subunits. For instance, the overall percentage and number of hydrophobic patches were calculated for the entire molecule, and a similar calculation was performed for the CDR loops region. In total, this team calculated approximately 70 protein biophysical and surface properties for each molecule. Patches were defined to be greater than one amino acid and used a proprietary surface area cutoff.

3.2 1A. Prediction of Protein A Elution Pool pH

For the prediction of protein A elution pool pH, the most successful approach titrated the MabSelect SuRe (resin) wash buffer with the elution buffer. The elution pHs of in-house mAb products were measured using the Highland Games experimental protocol and the data overlaid on the titration curve. The pH prediction was refined by adding an empirical adjustment to account for the impact of varying levels of acidic residues. The buffering capabilities of different mAbs are very comparable, and they are fairly dilute, so this potential contributing factor was excluded from the analysis.

The second most-successful approach was based on sequence alignment with mAbs from a proprietary database. Highland Games predictions used the experimentally-determined values of nearest neighbor sequences for proprietary mAbs.

The third most-successful approach assumed that the eluted protein “appeared” in the elution buffer, thereby disrupting the buffer’s equilibrium. This was achieved by using the equations described by Pabst and Carta (Pabst and

Carta, 2007) and calculating the solution pH using equations 1-3, while including the protein charge contribution (equations 4-6) to the electro-neutrality equation. The equations are solved iteratively, with pH as the only unknown (changes in the pH estimate will lead to changes in ionic strength, and therefore in protein charge).:

<i>Electro-Neutrality:</i>	$[H^+] + [Na^+] + [trisH^+] = [OH^-] + [Cl^-] + [AcO^-]$	(1)
<i>Equilibrium:</i>	$[AcO^-] = \frac{K_{ac}^{eff} C_{ac}}{K_{ac}^{eff} + [H^+]} / [trisH^+] = \frac{[H^+] C_{tr}}{K_{tr}^{eff} + [H^+]} /$ $[OH^-] = \frac{K_w}{[H^+]}$	(2)
<i>Mass Balance:</i>	$C_{ac} = [AcO^-] + [AcOH] / C_{tr} = [trisH^+] + [tris]$	(3)

Where Ka^{eff} is the effective equilibrium constant (*ac* for acetate, *tr* for tris) and C_{ac} and C_{tr} are the total concentrations of acetate and tris respectively. A slightly modified version of the Davies equation (Davies, 1938) is used to calculate the effective Ka^{eff} values as a function of ionic strength. In these equations, the only unknown (pH) is obtained iteratively. This system of equations can be expanded to include any buffering agent or charged species (succinates, phosphates, amino acids, etc.) in equations 4-6 and is used to calculate the pH and ionic strength:

Amino acids with positive residues (and N-terminus)	$\sigma_i = \frac{F \rho_i}{1 + K a_i^{eff} / [H^+] + K^{Cl} \gamma_{\pm NaCl} [Cl^-]}$	(4)
Amino acids with negative residues (and C-terminus)	$\sigma_i = \frac{F \rho_i}{1 + [H^+] / K a_i^{eff} + K^{Na} \gamma_{\pm NaCl} [Na^+]}$	(5)
Total Charge	$\sigma_p = \sum_i \sigma_i$	(6)

Where σ_i and σ_p are the charge densities of the different amino acid residues i and of the total protein respectively; F is the Faraday constant; ρ_i is the surface density of amino acid residue i ; $K a_i^{eff}$ is the effective equilibrium constants of the different amino acid residues (calculated as above); K^{Cl} and K^{Na} are the association constants between Cl^- and Na^+ ions and oppositely charged residues; and $\gamma_{\pm NaCl}$ is the mean activity coefficient of $NaCl$, calculated using Bromley's method (Zemaitis et al., 1986).

Some known limitations of this method, which are not expected to greatly alter the results, are that (a) the counterions inevitably present along with the protein are neglected, as they are very difficult to quantify and their concentration typically quite low; (b) the volume occupied by the protein in solution, which can affect ion concentration and pH, is neglected, as it is rather low (mAb at 30g/L would occupy ~2% of the volume of the solution); (c) the volumes of the elution

pools were themselves estimated in the context of the Highland Games, introducing high uncertainty to the protein concentrations values used for this calculation.

3.3 1B. Prediction of Protein A Elution Pool volume (CV)

The most successful approach to predicting protein A elution peak volume was based on outputs from homology modeling. The factors influencing Protein A retention are mainly electrostatic and hydrophobic interactions, and conformational flexibility of the mAb during elution. The outputs encompassed Hydrophobic moment, Total charge, Aggregation propensity (AggScore; Sankar et al., 2018). The team purified four in-house mAbs through the Highland Games experimental conditions and compared results against homology models of these in-house molecules. AggScore demonstrated the strongest correlation. Then AggScore from unknown mAbs were compared to in-house mAbs, and elution pool volumes were correlated.

The second-most-successful approach was based on sequence alignment, and followed the same protocol described for prediction of protein A elution pool pH. The third most-successful approach simply assumed the elution volume to be 2 column volumes; a typical elution volume encountered during in-house experiments.

3.4 Prediction of salt concentration for Capto Q elution

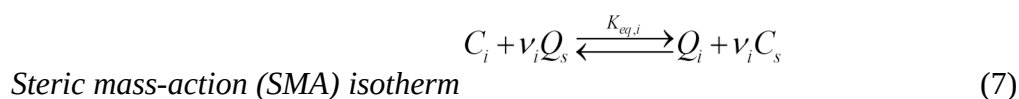
The most successful approach used was based on predictions using structural homology modeling and predictive models trained on prior data. Bio MOE software (Chemical Computing Group ULC, 2016) was used to convert antibody

sequences into full-length IgG homology models, with side-chain charges protonated to reflect pH at 24 different anion-exchange and 48 different cation-exchange operating conditions. From those structures, a complete set of MOE protein descriptors were calculated, including structural properties such as net charge, hydrophobic and electrostatic patches, and then exported to a descriptor database. A training set of data was created representing 36 IgG mAbs at each pH; the set consisted of their descriptors as well as lab-measured binding to anion- and cation-exchange resins. The degree of binding of mAb to resin was represented mathematically as Log Kp, the log of the partition coefficient (McDonald et al., 2016).

Construction of predictive models was done with SAS JMP (SAS Institute Inc., 2019). With structural descriptors as the input and Log Kp as the output, a partial least-squares method was used to construct one model for each of the 72 ion-exchange binding conditions. To generate predicted Log Kp values for Highland Games event mAbs, their sequences were used to calculate descriptors using the homology modeling method described above. Those descriptors were then used as input to the predictive models. For ease of analysis, these binding predictions were represented as 3D contour surface plots defined by points with coordinates: (sodium concentration, pH, predicted Log Kp). The composition of the elution buffer was predicted by interpolating these points to return the salt concentration at which Log Kp = 1.0 at pH 5.5 (cation exchange) or pH 8.0 (anion exchange).

The second-most-successful approach used the SMA model (Brooks and

Cramer, 1992) along with the elution predictions of Pedersen et al. (2003) which predicts the retention volume of a species eluted during a linear gradient as an offset from the retention volume of a non-retained species, as a function of the SMA characteristic charge (average number of binding sites interacting during adsorption), the ionic capacity of the adsorbent, the number of charges per counterion, the interstitial and intraparticle porosities, and the equilibrium constant and size exclusion factor. A characteristic charge is estimated using the average pKas of individual amino acids, and the equilibrium constant is estimated by comparison with a well-characterized in-house molecule:



$V_R = V_{NA} + 1 / G \left[GB \left(\frac{\Lambda}{z_s} \right)^\nu (\nu + 1) + C_{salt_0}^{\nu+1} \right]^{1/(\nu+1)} - \frac{c_{salt_0}}{G}$	(8)
$G = \frac{c_{salt_1} - c_{salt_0}}{V_g}$	(9)
$B = V_{Column} (1 - \epsilon) \epsilon_p K_d K$	(10)

Where, C_{salt_0} and C_{salt_1} represent the initial and final salt concentrations of the gradient respectively; V_g and V_{Column} are the gradient and column volumes; ϵ and ϵ_p represent the interstitial and intraparticle porosities; K_d is the size exclusion factor for Salt is assumed to be 1, and z_s is the number of charges per counterion

is assumed to be 1.

Two participants tied for the position of the third-most-successful approach. One adapted the methodology presented in Khalaf et al. (2016), which begins with the mass balance on a component i on a chromatography column using a lumped kinetic model described in equations 11 and 12:

Mobile-phase MB	$\frac{\partial c_i}{\partial t} = \frac{u_{sf} d_{ax,i}}{\varepsilon_i} \frac{\partial^2 c_i}{\partial x^2} - \frac{u_{sf}}{\varepsilon_i} \frac{\partial c_i}{\partial x} - \left(\frac{1 - \varepsilon_i}{\varepsilon_i} \right) \frac{\partial q_i}{\partial t}$	(11)
Solid-phase MB	$\frac{\partial q_i}{\partial t} = k_{m,i} (q_i^{eq} - q_i)$	(12)

Where c_i and q_i are the mobile and solid phase concentrations as a function of time t and distance along the column x ; $d_{ax,i}$, $k_{m,i}$, ε_i , and q_i^{eq} are the axial dispersion coefficient, the lumped mass transfer coefficient, the accessible porosity and the equilibrium solid phase concentration (each for the component i); and u_{sf} is the superficial velocity of the mobile phase. With differential equations 11 and 12 the concentration of any component (buffer, salt, protein) can be simulated at any position in the column and at any time. This means that if all the variables are known, then a chromatogram can be predicted, *a priori*. In fact, these variables can be estimated based only on the protein amino acid sequence, resin-specific descriptors, and operating conditions. With all the variables in equations 11 and 12 known, it is then possible to predict chromatograms for any protein on any ion exchange resin and at any operating/experimental conditions. From these

chromatograms, the elution sodium chloride concentration is obtained. It is important to note that this approach assumes a heterogeneous charge distribution on the considered protein. This assumption works well in general but struggles in cases where different parts of the protein are very oppositely charged. An example of this is mAb 1 at pH 8, the conditions for the Capto Q event. The pI of mAb1 is measured at 7.93. As such, no interaction should exist between this molecule and the Capto Q resin at pH 8. Adsorption is experimentally observed, however, and the elution sodium chloride is 52 mM. In this case, the light chain (pI ~ 5) is responsible for the binding: according to the model, a molecule of this charge and size would elute at ~60 mM sodium chloride. Unfortunately, to their regret, the team did not provide this value as a prediction! It relied on the full protein charge, and thus predicted no adsorption.

The other participant that tied for the third position in this event employed an approach based on QSAR Model Development. QSAR models were developed in Python using the scikit-learn library to predict Highland Games mAb CEX (Poros 50 HS) and AEX (Poros 50 HQ) retention as well as solution viscosity. For all models, the complete data set was randomly divided into training and test sets, with the test set excluded from model development and used for external validation in addition to internal cross-validation. Descriptors were standardized prior to model development. All three models (CEX, AEX, Viscosity) utilized the partial least squares (PLS) algorithm. The variable influence on projection (VIP) score was used for feature selection for both IEX models while recursive feature selection was used for the viscosity model due to the smaller number of

descriptors considered in this case.

For the ion-exchange models, existing partition coefficient K_p data from an internal process-development mAb database was converted into an equivalent elution peak first moment using a system offset. The offset was established separately for AEX and CEX by comparing high-throughput K_p data with benchtop linear gradient chromatography experiments according to the methodology of McDonald et al. (2016). For CEX model development, a limited subset of the internal process development mAb database that were known to exhibit anomalous experimental behavior were excluded from the training set. For the AEX model, mAbs that flowed through were omitted from model development and used as an additional validation step. The ability of the AEX model to correctly predict flowthrough for these mAbs was assessed in addition to the standard cross-validation and analysis of external test set predictions.

3.5 Prediction of salt concentration for POROS 50 HS elution

The top three most successful approaches to this event were the same as those for Event 2, Capto Q anion-exchange elution.

3.6 Prediction of viscosity (cP; concentration)

This event was impaired by the difficulty of concentrating almost all of the test mAbs to the target 160 g/L concentration (in some cases because of material limitations), precluding fair comparison of methods. Methods are therefore described, but not ranked in detail. The method which achieved the closest prediction of the viscosity of mAb 2 (at 156 g/L) began by noting that the low-shear viscosity η_0 of Newtonian colloidal systems where the colloids are non-

interacting spheres dispersed in a fluid of viscosity η_F can be described using the hard sphere model described in equation 13 (Berli et al., 2005):

$$\text{Hard Sphere viscosity} \quad \frac{\eta_0}{\eta_F} = \left(1 - \frac{\phi_s}{\phi_m}\right)^{-2} \quad (13)$$

Where ϕ_s is the volume fraction occupied by the colloid, and ϕ_m is the glass transition volume fraction ($\phi_m = 0.58$).

Proteins are of course not Newtonian non-interacting spheres (Buck et al., 2015; Fischer et al., 2015), but there are many literature examples where the low-shear viscous behavior of protein solutions resembles that of Newtonian hard spheres. The viscosity is not a function of the volume fraction, but of the apparent volume fraction (Nicoud et al., 2015; Sarangapani et al., 2013). The team developed a methodology to estimate the apparent volume fraction of protein solutions, considering three effects: the actual volume fraction of proteins in solution; the electro-viscous effect; and finally, network formation in high-concentration protein solutions. Equation 13 for protein solutions is then rewritten as:

Apparent viscosity	$\frac{\eta_0}{\eta_F} = \left(1 - \frac{\phi_{app}}{\phi_m}\right)^{-2}$	(14)
Apparent Volume Fraction	$\phi_{app} = c_i \frac{4}{3} \pi (a_s + a_{el} + a_{ne})^3$	(15)

Where c_i is the concentration of the protein solution; and a_s , a_{el} , and a_{ne} are the size, electro-viscous and network formation contributions to the radius of a sphere equivalent to the protein. The apparent volume fraction ϕ_{app} is then calculated using this equivalent sphere radius.

To estimate the contribution of the protein size a_s , the specific volume of the protein is calculated as described in Harpaz et al. (1994) and used in Equation 16 to obtain the size contribution:

Size Contribution to Viscosity

$$a_s = \sqrt[3]{\frac{3}{4} \frac{v_p M_w}{\pi N_a}} \quad (16)$$

Where v_p and M_w are the specific volume and the molecular weight of the considered protein, and N_a is Avogadro's number. The visco-electric contribution

Electro-viscous Contribution to Viscosity

$$a_{el} = \frac{1}{2} \int_{2a_s}^{\lambda} \left(1 - \exp \left(- \frac{z^2 l_b (2 \cosh(\kappa(R/2-b)) - (\kappa(R/2-b))^2 - 2)}{a_s (\kappa a_s \sinh(\kappa(b-a_s)))^2} \right) \right) dR \quad (17)$$

is calculated as described in equation 17 (Berli et al., 2005):

Where λ , l_b , κ , and b are the vanishing distance, the Bjerrum length, the inverse Debye length, and the cell radius. Finally, the network formation contribution is obtained through a method heavily adapted from that of Tomar et al. (2017). The obtained a_{ne} is a function of the solution pH and ionic strength, and of features in the surface amino acid sequences of the molecule considered.

Calculating the viscosity using these three contributions for a series of internal

molecules yields a dependence similar to that in Berli et al. (2005), Nicoud et al. (2015) and Sarangapani et al. (2013). It is important to note that a good understanding of the solution composition and pH is required. For all the examples analyzed internally, the UFDF model of Miao et al. (2009) was used to first predict the composition of the high-concentration protein solution. Prediction of viscosities for high concentration solutions generated by other methods has not been undertaken due to this limitation. Another approach used a QSAR model as described under Capto Q retention prediction to predict Highland Games mAb viscosity based on data available in the literature (Li et al., 2014), which included both viscosity measurements and descriptors. These additional features were also calculated for the Highland Games mAbs following the procedures outlined in Li et al. (2014) and used to generate the viscosity predictions.

A third approach used an in-house Coarse-Grained (CG) molecular modeling tool to predict the viscosity of concentrated mAb solutions. This CG modeling method represents the full-length antibodies using 10 CG beads placed in a Y-shaped arrangement and connected through rigid bonds. The point charges on each CG bead are assigned so that electrostatic molecular multipole moments, up to the fourth order, of the full-length antibody's charge distribution are optimally reproduced (Anandakrishnan et al., 2013). The short-range Lennard-Jones (LJ) potential parameters on the CG beads are adjusted to represent the hydrophobic character and radius of hydration of each representative CG domain. The CG model force field parameters were tuned using an internal set of 15 molecules with viscosity data available at high concentration, formulated in 20

mM histidine-acetate at pH 5.5. Applying the CG model on the same internal viscosity dataset showed that the self-diffusion coefficients extracted from CG simulations inversely correlate with measured viscosities at high concentrations. CG simulation of the six antibodies from the Highland Games dataset yielded self-diffusion coefficients. Linear regression between self-diffusion coefficients and measured viscosities obtained on an internal dataset was used to predict the 160 g/L viscosities of the Highland Games antibodies from their calculated self-diffusion coefficients.

In more detail, the Coarse-Grained simulation protocol used was as follows. First, full-length antibody homology models (aglycosylated) were built and protonated at pH 6.0. CG models were built based on the electrostatic and hydrophobic characteristics of each molecule as described above. A simulation box of 1300 Å was then filled by 1460 full-length antibody CG models, representative of 160 g/L protein concentration. Periodic boundary conditions were applied in all three directions. The CG simulations were performed under NVT conditions using Langevin dynamics simulation protocol in the LAMMPS package [3]. A Langevin thermostat was used to set the temperature to 300 K. The CG simulations for the rigid bodies were run for 5 microseconds, using a timestep of 1 ps. The first two microseconds were excluded, and the self-diffusion coefficient for each antibody CG model was calculated on 3-microsecond trajectories using CPPTRAJ software (Roe and Cheatham, 2013) available in AmberTools. Three independent simulations were conducted using the protocol described above, and the self-diffusion coefficients from three simulations were

averaged.

Another team's predictions were based on the work of Sharma et al. (2014), who correlated properties of 60 IgG₁ antibodies with various characteristics and found that viscosity increases strongly with hydrophobicity and charge dipole distribution, and decreases with net charge. Principal component regression analysis was used to develop a predictive model for viscosity. In-house data were used to refine the coefficients of the regression equation of, particularly by calculation of the charge distribution properties of the Fv fragment of the antibody. This model was further validated using a spatial charge map (SCM) tool developed by Agrawal et al. (2016), which provides an orthogonal high throughput sequence-based approach to viscosity prediction through the quantification of electronegative patches on the Fv domain of the antibody.

3.7 Prediction of aggregation stability at pH 3.4

This event was not scored because the stability of the tested mAbs was higher than expected. One team used their in-house data to predict that there would be no significant aggregation over 24 h (expressed in % HMW formed per day). Another team used a nearest-neighbor sequence alignment method with their internal mAb database. Some mAbs showed an increase in high molecular weight (HMW) species, whereas others showed no trend. Only Highland Games mAbs with nearest neighbors that had positive HMW rates were predicted to have an increase in HMW.

3.8 Prediction of acidic species formation rate at 25°C, pH 8.2, measured over 16 days (% per day)

The most successful approach noted that deamidation of asparagine (Asn) residues in the CDR region has been widely reported as a major cause of formation of acidic species (Harris et al., 2001; Lyubarskaya et al., 2006; Vlasak et al., 2009). Other possible contributors to the formation of acidic species include disulfide linkages (Martinez et al., 2008; Pristatsky et al., 2009), sialic acid (Khawli et al.,

2010; Lyubarskaya et al., 2006), and glycation (Khawli et al., 2010), to name a few. These other modifications are not as widely observed as deamidation, however, and were therefore not addressed in this work. In addition, deamidation of Asn residues in the constant regions was not considered because the molecules studied here are very similar in the constant region (IgG₁ framework), and thus these deamidation sites do not affect the final rank ordering and predicted values. The team focused, therefore, on predicting the number of deamidation sites as well as the deamidation rates for each Asn in the CDRs.

Deamidation of Asn starts with a nucleophilic attack of the residue's backbone nitrogen atom on the Asn's side chain amide group carbon atom to form an intermediate cyclic succinimide residue, which can further undergo hydrolysis to produce an aspartic acid (Asp) or isoaspartic acid (IsoAsp) (Wakankar and Borchardt, 2006). This reaction requires the Asn side chain to adopt a conformation that minimizes the nucleophilic attack distance between Asn's side chain amide group carbon atom and residue's backbone nitrogen atom. A 2.5 microsecond explicit-solvent Molecular Dynamics (MD) simulation of each Fab was conducted to characterize the propensity of the Asn side-chains in the CDR to be near a nucleophilic attack conformation. An internal dataset of 21 antibodies with variable CDR deamidation rates were used to identify side-chain conformations favorable for the reaction. The probability of deamidation for each potential site was scaled based on the identity of the N+1 residue (e.g., Gly, Ser, and Thr are considered to be more reactive). The predicted deamidation rates were then converted into % acidic species by fitting to in-house data.

The second-most-successful approach for prediction of acidic species formation at high pH used a combination of nearest-neighbor sequence alignment, prior literature values, and MOE model data. All predictions were based on asparagine deamidation liability. Highland Games mAb rates were initially predicted using the nearest neighbor-sequence and an internal mAb database. Rates for the Fc region were taken as a constant, in line with rates published by Pace et al. (2013). Contributions from liable asparagines in the Fab region were used to determine an overall rate. Predicted rates of Highland Games mAbs were adjusted for Fab asparagines using the deamidation coefficient descriptor in MOE models as described by Robinson (2002) and Capasso (2000).

In the third-most-successful approach, the crystal structure of human IgG₁ B12 (PDB ID: 1HZH) was used as the Fc template to create a full-length homology model of each Highland Games mAb in Schrödinger BioLuminate (Zhu 2014, Jacobson 2004). Briefly, the Antibody Prediction tool was used to identify the most similar Fv template in the PDB and graft CDR loops from a separate loop database. The Protein Preparation Wizard tool was used to add hydrogens, assign bond orders, optimize H-bond assignments, and perform restrained energy minimization. Deamidation half-lives were calculated following the method of Robinson and Robinson (2001b). Briefly, the structural parameters C_m , C_{Sn} , and S_n of each asparagine were calculated from the three-dimensional models. Experimental pentapeptide half-lives ($t_{1/2}$) were obtained from Robinson and Robinson (2001a). The predicted half-life for each asparagine was calculated as described by these workers, but half-life predictions at 25°C were obtained by

augmenting the half-lives from the above paper by a factor of 10.7/3.4, following the observations of Pace et al. (2012).

3.9 Prediction of isoelectric point

Competitors generally did very well at predicting mAb isoelectric points. The most successful approach used a combination of theoretical calculation and experimental calibration. Initial pI was determined using two algorithms implemented in Molecular Operating Environment (MOE, Chemical Computing Group): (1) sequence-based, and (2) coordinate-based. The methodology used for the sequence-based algorithm was that proposed by Meireles Ribeiro and Sillero (1991). This approach was modified to develop an empirical coordinate-based algorithm which accounts for perturbation of pKa values by local molecular environment effects (PROPKA algorithm, which includes effects due to desolvation, hydrogen bonding, and charge-charge interactions; Li et al., 2005). The coordinate-based pI prediction approach used homology models of entire mAbs, built as follows: (1) a homology model of the Fab region was built in MOE if a crystal structure of the Fab was not available; where multiple Fab crystal structures were available apo Fab structures with the highest resolution were chosen, (2) the Fab model was grafted onto the crystal structure of mAb 1HZH if the mAb of interest was an IgG₁ and mAb 5DK3 if the mAb of interest was an IgG₄, and (3) residues differing between the Fc of 1HZH or 5DK3 and that of the mAb of interest were mutated using the MOE Protein Builder module.

While the sequence-based algorithm will fail to account for the impact on pKa's of the local environment, the structure-based algorithm will account for this

impact but is dependent on the specific energy calculations used to calculate the effect of the local environment. Once the initial pI values of several experimental monoclonal antibodies in internal development were calculated using the two algorithms described above, the predicted values were plotted against experimental pI values, and a linear regression was performed. The sequence-based algorithm most reliably predicted the experimental pI values based on the regression statistics and was used for further prediction of pI values for mAbs of interest in the competition.

The second-most-successful approach used the H++ server (Anandakrishnan et al., 2012; Gordon et al., 2005; Myers et al., 2006) to calculate the isoelectric point for full-length antibody homology models. The methodology behind the H++ server is based on classical continuum electrostatics and basic statistical mechanics. H++ provides estimates of biomolecule characteristics such as pKas, isoelectric points, titration curves, and energies of protonation microstates.

The third-most-successful approach was as follows: with the pH and ionic strength known, the charge of the protein is calculated (at infinite dilution) based on the number of surface-exposed amino acids. This is done using modified forms of equations reported elsewhere (Guélat et al., 2010; Guélat et al., 2012; Guélat et al., 2016; Khalaf et al., 2016).

3.10 8A. Prediction of aggregation to High MW species (HMW) over 32 days (% per day)

The most successful approach utilized rate equations for stability prediction at 37 °C. Nearest-neighbor sequence alignment from an internal mAb database was used to predict the formation of HMW and low molecular weight (LMW) species during 37 °C stability studies. In cases where the concentration of nearest neighbor internal mAb stability studies differed from the 100 g/L concentration specified for the Highland Games, 2nd order reaction kinetics were used to estimate HMW formation according to Weiss et al. (2009).

The two teams which tied for second both constructed homology models, then used them to calculate AggScore (Sankar et al., 2018), a structure-based aggregation propensity score based on the intensity and distribution of hydrophobic and charged patches on the surface of the mAbs. AggScore was used instead of other sequence-based predictors since it has been shown to be significantly better at discriminating between the retention times of clinical-stage antibodies in various hydrophobic chromatographic columns and has been found by one team to correlate well with internal IgG₁ aggregation data. AggScore calculations were performed using the protein-specific descriptor calculation module in BioLuminate at pH 3.4 and 5.8. The AggScore values were then converted into % high molecular weight species by fitting to custom in-house data or interpolated based on an in-house data set. All calculations were performed using tools available in Schrodinger's BioLuminate package release 2018-2.

3.11 8B. Prediction of fragmentation over 32 days (% per day)

Two participants tied for the first position. One used the same nearest-neighbor approach as event 8A, and reported LMW formation was estimated to have a rate constant independent of mAb concentration. The other began with the assumption that the formation of low molecular weight species is primarily driven by fragmentation, i.e., disruption of a covalent bond in a protein by spontaneous or enzymatic reaction (Vlasak and Ionescu, 2011). Because mAb molecules share considerable sequence similarities, fragmentation rates are expected to be very similar for molecules in the same subclass (e.g., IgG₁). This team's historical data show that the amount of fragmentation is very small for mAb molecules in the IgG₁ framework, so they predicted a negligible fragmentation rate for the six Highland Games molecules. As a sanity check for the data-driven fragment prediction, they evaluated the probability of an asparagine-proline peptide bond cleavage, which can contribute to fragmentation in the CDR. These Asn-Pro cleavage predictions did not identify any cleavage-labile sites among the Highland Games mAbs. The second-most-successful approach was based on participant's in-house data suggesting an average fragmentation rate of 2%.

4. Results

Experimental results and predictions are summarized in Table 1, which lists the experimental average and range, and the average (absolute value of) prediction errors. Table 2.1 and 2.2 present all experimental and prediction results for each antibody and event. Along with the mAb sequences in SI Table 1, these

experimental results constitute a unique public resource for benchmarking and methods development. Figure 1 presents the results graphically, with individual teams represented by different (anonymized) symbols. This figure describes the scatter of data, the proximity to the experimental data and how different approaches can lead to different outcomes. More detailed statistical analysis for each individual challenge is not appropriate because the number of data points is low, it differs among different predictions and different methodologies were used to achieve the predictions. Figure 2 presents viscosity of mAbs as a function of concentration, to illustrate the remarkable range of behavior among the test set, and to provide access to detailed data for use in future development of prediction tools.

5. Discussion

The Highland Games competition draws upon a rich tradition of collaboration and benchmarking in the pre-competitive advancement of industries, including the elucidation of CHO cell metabolism (Hefzi et al., 2016), the benchmarking of MHC Class II binding prediction methods (Andreatta et al., 2018), the characterization of the NISTmAb (Schiel and Turner, 2018), the A-Mab QBD exercise (Lim et al., 2018), high-throughput process development (HTPD) Olympiad 2014 (www.htpdmeetings.com; Welsh et al., 2016) the Industrial Fluid Properties Simulation Challenge (<http://fluidproperties.org/>), the long-running CASP protein competitions of the Protein Structure Prediction Center (<http://predictioncenter.org/casp13>) and (in a different but increasingly-relevant field) the Kaggle competition (www.kaggle.com) for development of data science

tools. In all these diverse areas, competition and benchmarking has advanced the state-of-the-art, especially when competitors generously share knowledge of best practices. Highland Games' value has been greatly enhanced by such sharing.

Some events did not work well in this first iteration. The competition to predict mAb viscosity at 160 g/L was hindered by the inability to concentrate most antibodies to this concentration, though the few feasible prediction/experiment comparisons and the sometimes-large spread of the predictions suggests that viscosity prediction remains challenging. Most mAbs tested were not retained under the anion-exchange conditions used, and they also were unexpectedly stable against proteolysis and aggregation. In some cases, instability could not be meaningfully measured over the time specified. Potential future iterations of Highland Games will correct for these problems, and a call will be made for somewhat less-stable antibodies.

Based on this first iteration of Highland Games, industry capabilities vary widely across challenges. The prediction of protein A elution pH was quite successful, though it should be noted that most mAbs had similar Fc sequences. The prediction of Protein A elution pool volume was somewhat less reliable, with an experimental outlier being challenging for the prediction teams. Many mAbs did not bind under the conditions pre-selected for anion-exchange chromatography, but most teams were able to predict qualitative AEX binding. In comparison, the quantitative binding predictions for cation-exchange chromatography behavior were in some cases rather good. The selected target concentration for viscosity prediction (160 g/L) was too high to be reached with

most mAbs, but viscosity prediction clearly remains a challenge. Low rates of pH 3.4 aggregation, fragmentation, and HMW formation affected these events, possibly because the mature mAbs used in the competition have been selected for stability along with their development history. The prediction of acidic species formation was relatively successful, and prediction of pI was both popular and very successful. It should be noted that unknown minor variations in preparation and test conditions among companies also may have introduced unpredictable variations.

Sometimes simple heuristics were very effective. The third most-successful approach to predicting Protein A elution volume assumed the elution volume to be 2 column volumes; a typical elution volume encountered during in-house experiments. Despite the repeated success of empirical approaches, it is noteworthy that the majority of discussion at the Recovery of Biological Products Conference supported the idea that substantial and broadly-applicable improvements in prediction will be achieved through molecular modelling-based approaches.

Contestants were very generous in sharing their expertise, and the Methods section of this paper should prove valuable to the advancement of the field, and the development of prediction capabilities in new organizations. Much credit is due to these contestants and their organizations and management. The amino acid sequences and experimental characterizations of the contest mAbs allow for many further uses of the dataset, including internal self-testing. Some areas of prediction remain unsolved problems. The remarkable stability of some contest antibodies

means some prediction approaches await antibodies sufficiently unstable to really test their capabilities; future iterations of Highland Games should consider less-stable molecules. Finally, the community owes a considerable debt to the volunteers who performed the extensive experimental characterizations upon which the competition was based.

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Conflict of interest statement

No conflict of interest has been declared by the author(s).

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Figures

Figure 1: Experimental vs. predicted values for various events. The different colors represent different teams' predictions. Dashed lines represent perfect agreement between predicted and experimental values. Dotted lines are the regression lines of average prediction for each mAb as a function of its experimental value. Open circles show the experimental values for the six mAbs.

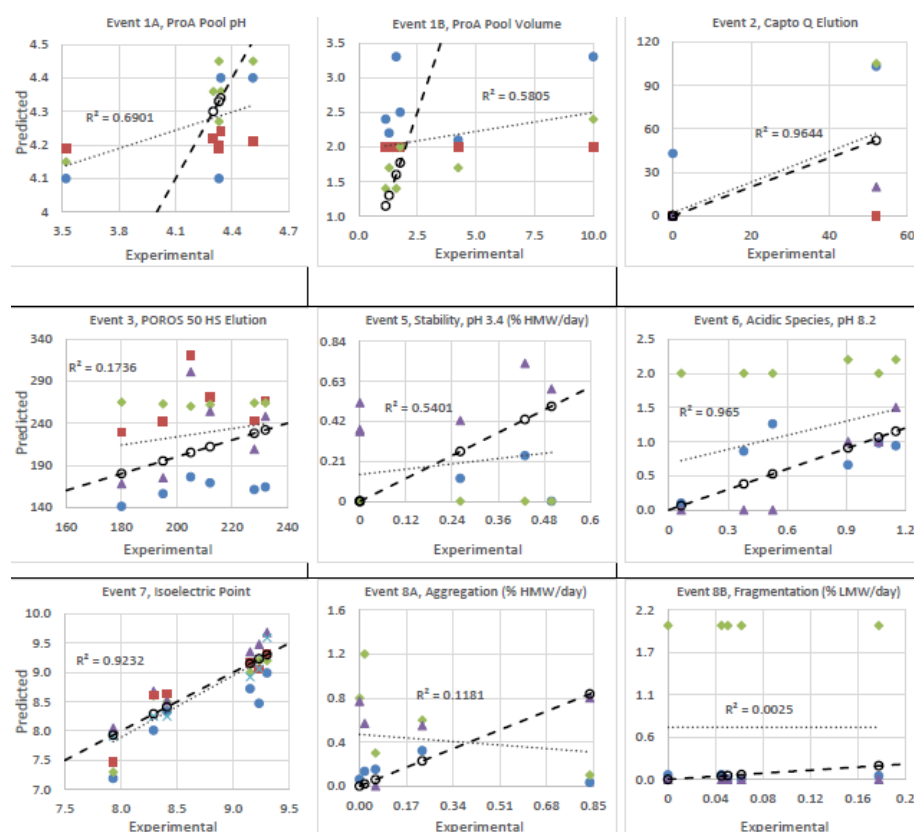


Figure 2: Concentration dependence of measured viscosity (cP) as a function of concentration (g/L) for four Highland Games mAbs. Viscosity measurements for (A) mAb 2 (red circles) and 6 (blue diamonds); (B) mAb 1 (black squares) and mAb 4 (orange triangles). Note that due to materials limitations and high viscosity, only mAb 2 could reasonably be concentrated to the target concentration of 160 g/L, and that mAb 4 showed very high viscosity even at its (low) maximum attained concentration of 48 g/L.

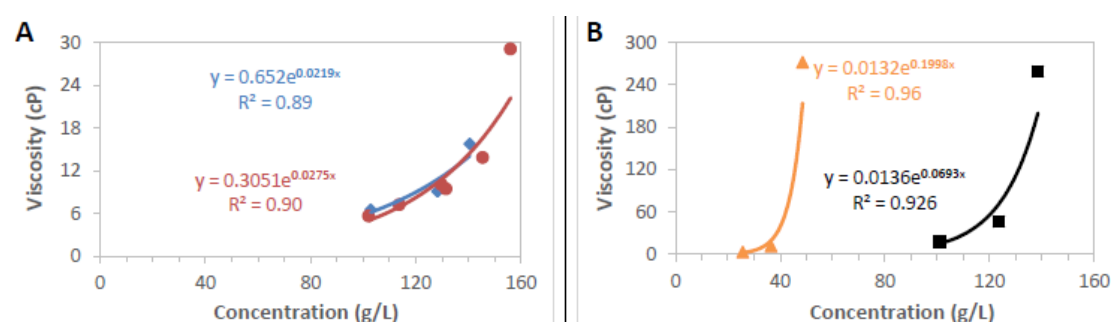


Table 1. Global summary of Highland Games results

Event	Experimental Average; Range	All-prediction Average Error	Most Successful Method; Empirical (E) or Modelling inputs (M) or both (E/M)	Comments
1A) MabSelect Sure Elution (pH)	4.2; 3.5-4.5	0.2	E/M	Excellent results; tight experimental range, likely due to

				similar mAb Fc's
1B) MabSelect Sure Elution Pool Volume (Col. Vol.)	3.3; 1.15-10	2.1	E/M	Similar mAb Fc considerations as event 1A. Single variant mAb challenging for prediction teams
2) Capto Q Elution Peak ([NaCl], mM)	8.7; 0-52	9.6	E/M	Only 1 of 6 mAbs bound under specified conditions, most teams able to predict binding in yes/no fashion
3) POROS 50 HS Elution Position ([NaCl], in mM)	208; 180-232	47.4	E/M	Fairly good prediction results
4) Viscosity (cP)	103 (N=2); 15.8-272	83.9	Not ranked	Only 1 mAb could be prepared at 160 g/L (viscosity, material limitations).
5) Aggregation at pH 3.4 (% HMW per day)	NA	0.2	N/A	No data points above limit of quantitation (0.5%). Industry selection for very stable mAbs?
6) Acidic Species Formation (%)	0.6; 0.06-1.15	0.7	E	Challenging event with wide range of experimental results. Fair prediction

<i>per day)</i>				performance
7) Isoelectric Point [pI]	8.8; 7.93-9.30	1.6	E/M	Only event with submissions from all prediction teams. Excellent results indicate maturity of this discipline
8A) Long-term Stability, HMW (% <i>per day</i>)	0.2; 0-0.837	0.5	M	Low levels of aggregation made scoring and differentiation difficult. Industry favors stable mAbs?
8B) Long-term Stability, Fragmentation (% <i>per day</i>)	0.1; 0-0.177	0.7	E; E/M (tied)	Relatively low levels of fragmentation impaired scoring, ranking. Selection for stable mAbs?

Table 2.1: Experimental and predicted values for all events for mAb1, mAb2, and mAb3

Event #; Description	mAb1		mAb2		mAb3	
	Exp.	Pre d.	Exp.	Pre d.	Exp.	Pre d.
1A) MabSelect Sure Elution (pH)	4.33	4.1, 4.2, 4.45	4.34	4.24, 4.36, 4.4	4.51	4.21, 4.4, 4.45

1B) MabSelect Sure Elution Pool Volume (Column Volumes)	1.6	2, 1.4, 3.3	1.5	1.7, 2, 2.1	1.1 5	2, 1.4, 2.4
2) Capto Q Elution Peak Position ([NaCl], in mM)	52	0, 20, 103, 105	0	0, 0, 0, 0	0	0, 0, 0, 0
3) POROS 50 HS Elution Position ([NaCl], in mM)	180	141, 168, 229, 265	205	176, 260, 301, 321,	232	164, 248, 264, 267
4) Viscosity (cP)	259 cP @ 138 g/L	21.6 , 34.1 , 39.2 , 74.4	29.1 cP @ 156 g/L	6.5, 12.9 , 13.4 , 20.4	18. 2 cP @ 145 g/L	2.3, 13.5 , 22.3 , 122
5) Aggregation at pH 3.4 (% HMW per day)	0.26 16	0, 0.12 , 0.42 5	0	0, 0.37 7	0.5	0, 0, 0.59 0
6) Acidic Species Formation (% per day)	0.52 7	0, 1.26 , 2	1.15	0.94 , 1.5, 2.2	1.0 6	0.98 , 1, 2
7) Isoelectric Point [pI]	7.93	7.19 , 7.3, 7.47	9.3	8.99 , 9.2, 9.32	9.2 3	8.47 , 9.05 ,

		, 7.91 , 8.05		, 9.59 , 9.68		9.06 , 9.2, 9.48
8A) Long-term Stability, HMW (% per day)	0.05 8	0, 0.15 2, 0.3	0.22 79	0.32 1, 0.55 , 0.6	0.8 37	0.03 2, 0.1, 0.8
8B) Long-term Stability, Fragmentation (% per day)	0.04 5	0, 0.06 1, 2	0.17 74	0, 0.04 5, 2	0	0, 0.05 9, 2
Exp. and Pred. denote experimental and predicted values, respectively.						

Table 2.2: Experimental and pooled predicted values for different events for mAb5, and mAb6

Event #; Description	mAb4		mAb5		mAb6	
	Ex p.	Pre d.	Exp .	Pred.	Exp .	Pred.
1A) MabSelect Sure Elution (pH)	3.5 2	4.1, 4.15 , 4.19	4.33	4.19, 4.2, 4.27	4.3	4.22, 4.3, 4.36
1B) MabSelect Sure Elution Pool Volume (Column Volumes)	10	2, 2.4, 3.3	1.77	2, 2, 2.5	1.3	1.7, 2, 2.2

2) Capto Q Elution Peak Position ([NaCl], in mM)	0	0, 0, 0, 43	0	0, 0, 0, 0	0	0, 0, 0, 0
3) POROS 50 HS Elution Position ([NaCl], in mM)	195	156, 175, 242, 263	228	161,2 09, 243, 264	212	169,2 54, 262, 271
4) Viscosity (cP)	272 .1 @ 48. 5 g/L	17.7 , 27.8 , 92.4 , 93.8 ,	23.9 cP @ 118 g/L	27.2, 40.7, 41.7, 73.7	15.8 cP @ 141 g/L	17.5, 17.8, 19.2, 20.2
5) Aggregation at pH 3.4 (% HMW per day)	0.4 30	0, 0.24 , 0.72 48	0	0, 0, 0.367	0	0, 0, 0.518
6) Acidic Species Formation (% per day)	0.3 8	0, 0.86 , 2	0.90 8	0.66, 1, 2.2	0.06 29	0, 0.1, 2
7) Isoelectric Point [pI]	8.2 9	8.01 , 8.25 , 8.6, 8.61 , 8.68 ,	8.41	8.25, 8.33, 8.5, 8.51, 8.63	9.15	8.72, 8.92, 9, 9.16, 9.35,

8A) Long-term Stability, HMW (% per day)	N/A	0.25 8, 1, 1.8	0.01 87	0.57, 1.2, 0.133	0	0.061 , 0.77, 0.8
8B) Long-term Stability, Fragmentation (% per day)	N/A	0, 0.04 , 2	0.05 03	0, 0.004 , 2	0.06 18	0, 0.002 , 2

N/A: Not Available, limited material availability.